

Mechanistic Modeling of the Loss of Protein Sieving Due to Internal and External Fouling of Microfilters

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Fed-batch and perfusion cell culture processes used to produce therapeutic proteins can use microfilters for product harvest. In this study, new explicit mathematical models of sieving loss due to internal membrane fouling, external membrane fouling, or a combination of the two were generated. The models accounted for membrane and cake structures and hindered solute transport. Internal membrane fouling was assumed to occur due to the accumulation of foulant on either membrane pore walls (pore-retention model) or membrane fibers (fiber-retention model). External cake fouling was assumed to occur either by the growth of a single incompressible cake layer (cake-growth) or by the accumulation of a number of independent cake layers (cake-series). The pore-retention model was combined with either the cake-series or cake-growth models to obtain models that describe internal and external fouling occurring either simultaneously or sequentially. The models were tested using well-documented sieving decline data available in the literature. The sequential pore-retention followed by cake-growth model provided a good fit of sieving decline data during beer microfiltration. The cake-series and cake-growth models provided good fits of sieving decline data during the microfiltration of a perfusion cell culture. The new models provide insights into the mechanisms of fouling that result in the loss of product sieving. © 2017 American Institute of Chemical Engineers Biotechnol. Prog., 33:1323–1333, 2017

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Introduction

Most antibodies are currently produced in batch or fed-batch cell culture bioreactors. However, perfusion processes have a number of advantages. A perfusion operation provides constant nutrient addition and metabolite removal allowing for higher cell densities and product titers than can be typically achieved. This has the potential to reduce facility footprint and allow increased use of disposable equipment.¹ Both fed-batch and perfusion cell culture processes can use microfilters (MF) for product harvest. Conventional tangential flow filtration (TFF) operations continuously pump feed from the bioreactor across a filter channel and back to the bioreactor while cell-free permeate is drawn off and collected. Alternating tangential flow (ATF) systems use an alternating flow diaphragm pump that pulls and pushes feed from and to the bioreactor while cell free permeate is drawn off. The ATF systems typically have less shear stress and smaller equipment than TFF operations.

Both TFF and ATF operations can be limited by membrane fouling, which is detected by pressure increase at constant flow, flow rate decrease at constant pressure, or by loss of

product sieving. Variable or declining product sieving can impair process robustness and economics. A number of studies have experimentally and theoretically investigated the effects of MF fouling on membrane resistance and protein sieving.

Blanpain-Avet et al.² provided data showing a resistance increase and sieving loss during the microfiltration of clarified beer using a membrane with straight cylindrical pores in a stirred cell. The flow resistance increase was modeled using the standard and cake blocking models. The sieving decline was not modeled.

Jonsson et al.³ provided data for resistance increase during the filtration of bovine serum albumin (BSA) with polysulfone (PS), polyethersulfone (PES), and polypropylene microfiltration membranes in a TFF device. The resistance data followed the intermediate-blocking model initially and subsequently the cake-blocking model.

Filippov and Starov provided data for flux decay during the filtration of colloid-rich tap water with track-etched polyethylene terephthalate and Teflon microfiltration membranes⁴ in a dead-ended membrane holder. Models of sieving decline and resistance increase were derived assuming the pore size distribution shifted as a result of fouling.

Kelly et al. provided data for pressure increase during the ATF filtration of a PER.C6 cell culture bioreactor with

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hollow fiber PES microfiltration membranes.⁴ The data were modeled with modified forms of the standard and cake blocking models.

Hadpe et al.⁵ provided data for pressure increase and flux decay during the ATF filtration of a CHO cell culture bioreactor with hollow fiber PES microfiltration membranes. The data were modeled with modified forms of the classic blocking models.

Ho and Zydney⁶ provided data for flux decay during the stirred cell microfiltration of a BSA solution through track-etched polycarbonate membranes. The data were modeled assuming fouling occurred initially due to pore blockage and subsequently due to cake formation.

Palacio et al.⁷ provided data for flux decay during the stirred cell microfiltration of BSA, lysozyme, pepsin, immunoglobulin G, and myoglobin through track-etched polycarbonate membranes. The data were fit using the models of Ho and Zydney.

Guell and Davis⁸ provided data for resistance increase and sieving loss during the filtration of a mixture of BSA, lysozyme, and ovalbumin with four different microfiltration membranes in a stirred cell. Kuberkar provided data for resistance increase during the filtration of BSA with yeast cells using cellulose acetate membranes in a TFF device.⁹ Clink et al. provided data for pressure increase and sieving loss during the TFF or ATF filtration of an IgG producing CHO cell culture bioreactor with hollow fiber PS microfiltration membranes.^{10,11}

However, there have been limited efforts to theoretically model the loss of sieving with fouling. Persson et al. provided data for flux and sieving decline during the filtration of BSA with Nylon and PS membranes in a TFF device. A model of sieving decline was obtained by combining the standard model of membrane fouling due to pore constriction with an empirical model of sieving as a function of pore radius.¹²

In this study, models of sieving loss due to internal membrane fouling, external membrane fouling, or a combination of the two were generated. Literature data that provided both sieving and membrane resistance as a function of time or volume were used to test the models. The models accurately described the loss of protein sieving during beer microfiltration and the loss of antibody sieving during TFF perfusion of a bioreactor.

Modeling of Sieving Decline Due to Fouling: Internal Membrane Fouling

Internal fouling of membranes composed of straight cylindrical pores

Theoretical modeling of sieving decline as a function of membrane fouling has been limited. To generate a physically accurate model, previous efforts to model sieving loss or resistance increase were investigated.

Membrane resistance increase has typically been described to occur due to one of three physical phenomena either alone or in combination. In standard blocking, membranes have straight cylindrical pores that decline in radius as solid matter accumulates on the pore walls. Complete or intermediate-blocking describes membrane resistance increasing due to foulants blocking pore entrances. Cake fouling describes resistance increase due to the buildup of cake layers on top of the membranes.¹³ The fouling of a

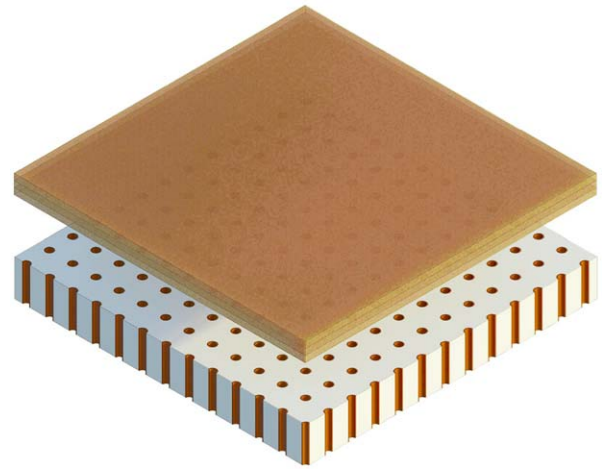


Figure 1. Illustration of a cylindrical pore membrane (grey) fouling due to the combined effects of caking and standard blocking.

Pores are constricted by solid matter (tan) accumulating evenly along the pore walls. In addition, a cake layer (tan) of increasing thickness accumulates above the membrane.

cylindrical pore membrane due to the simultaneous effects of standard blocking and cake fouling is illustrated in Figure 1.

Persson et al. developed a model by combining the standard model of membrane fouling due to pore constriction with an empirical model of sieving as a function of pore radius.¹²

Standard blocking has been described previously.¹⁴ A membrane with a thickness L_M , pore radius r , and pore density N will undergo a reduction in pore radius due to the filtration of a normalized volume V ($\text{m}^3 \text{m}^{-2}$) of a fouling liquid. Here c is the volume concentration of foulant and ϵ is the porosity of the foulant deposited on the membrane walls. The subscript 0 denotes initial parameter values.

$$-2\pi N L_M r dr = \frac{c}{1-\epsilon} dV \quad (1)$$

Integrating Eq. 1 yields

$$\pi N L_M (r_0^2 - r^2) = \frac{c}{1-\epsilon} V \quad (2)$$

The maximum filtration volume V_{Max} will be obtained when the pore radius r declines to zero due to the accumulation of foulants. Based on Eq. 2 when $r = 0$, $V_{\text{Max}} = (1-\epsilon)\pi N L_M r_0^2 / c$. Rearranging Eq. 2 and putting it in terms of V_{Max} yields

$$\frac{r}{r_0} = \sqrt{1 - V/V_{\text{Max}}} \quad (3)$$

Using Eq. 3 allows calculation of λ , the ratio of the solute radius r_s to the pore radius r , as a function of volume

$$\lambda = \frac{r_s}{r} = \frac{r_s}{r_0 \sqrt{1 - V/V_{\text{Max}}}} \quad (4)$$

To derive an empirical model of sieving as a function of volume, Eq. 4 is inserted in the empirical model shown in Eq. 2 of Persson et al. (Eq. 5 in this study) and then rearranged in terms of the initial sieving value S_0

$$S = \left(1 - \frac{r_s}{r}\right)^2 = \left(1 - \frac{r_s}{r_0 \sqrt{1 - \frac{V}{V_{\text{Max}}}}}\right)^2 = \left(1 - \frac{1 - \sqrt{S_0}}{\sqrt{1 - \frac{V}{V_{\text{Max}}}}}\right)^2 \quad (5)$$

Filippov et al.⁴ derived a model of the sieving of different sized particles through membrane pores with a log-normal distribution of sizes. Particles with radii below those of the pores were assumed to sieve freely. Sieving was predicted to decline with filtration volume as pores became plugged, shifting the pore size distribution. The model did not consider other physical mechanisms including the reduction of pore diameter due to fouling on pore walls, the partitioning of particles into pores, and the rates of convection and diffusion within pores. The four-parameter model was solved numerically.

To improve on the empirical model of Persson et al. and the numerical model of Filippov et al., the physical phenomenon underlying sieving and membrane fouling were considered. A number of equations have been developed to describe the convective and diffusive contributions to the sieving of a spherical solute through a straight cylindrical pore¹⁵ based on λ . The sieving coefficient S depends on the overall hindrance factor for convection W , the overall hindrance factor diffusion H , the protein free solution diffusivity, D , filter thickness L_M , and the solvent velocity J

$$S = \frac{W}{1 - (1 - W) \exp\left(-\frac{WL_M J}{HD}\right)} \quad (6)$$

where the overall convective hindrance factor incorporates pore partitioning effects. The overall diffusive and convective hindrance coefficients H and W can be calculated from^{15,16}

$$W = (1 - \lambda)^2 \left(\frac{1 + 3.867\lambda - 1.907\lambda^2 - .834\lambda^3}{1 + 1.867\lambda - .741\lambda^2} \right) \quad (7)$$

$$H = 1 + \frac{9}{8} \lambda \ln \lambda - 1.560\lambda + .528\lambda^2 + 1.915\lambda^3 - 2.819\lambda^4 + .270\lambda^5 + 1.101\lambda^6 - .436\lambda^7 \quad (8)$$

where Eq. 7 is valid for $0 \leq \lambda \leq 0.95$ and Eq. 8 is valid for $0 \leq \lambda \leq 1$. To generate a model of the decline in sieving due to membrane fouling, the sieving equation was combined with the standard model of membrane fouling due to pore constriction by inserting Eq. 4 for λ into Eqs. 7 and 8 above.

Inserting Eq. 4 into Eqs. 6, 7, and 8 provides the full pore-retention model, which describes sieving as a function of volume filtered. The term r_s/r_0 can be determined when $V = 0$ by inserting $\lambda = r_s/r_0$ into Eqs. 6, 7, and 8 and finding the value that best predicts the measured initial sieving value S_0 . The term V_{Max} can be determined by fitting the sieving versus volume data. Alternatively, the V_{Max} value derived from the pressure increase at constant flow, or flow decrease at constant pressure can be used. This model allows extrapolation of limited sieving data to predict sieving at higher volumes.

A number of simplifications can be made to the sieving model. If it is assumed that the diffusive contribution to sieving is low and the value of H is near zero, which is typical for proteins, than Eq. 6 simplifies to $S = W$. Equation 7 for W can be approximated by the following linear equation, which is applicable for $0 \leq \lambda \leq 0.95$

$$S = 1 - 1.17\lambda \quad (9)$$

A comparison of Eq. 7 and Eq. 9 is provided in Figure 2.

Substituting Eq. 4 into Eq. 9 and rearranging in terms of the initial sieving value S_0 yields the approximate pore retention model

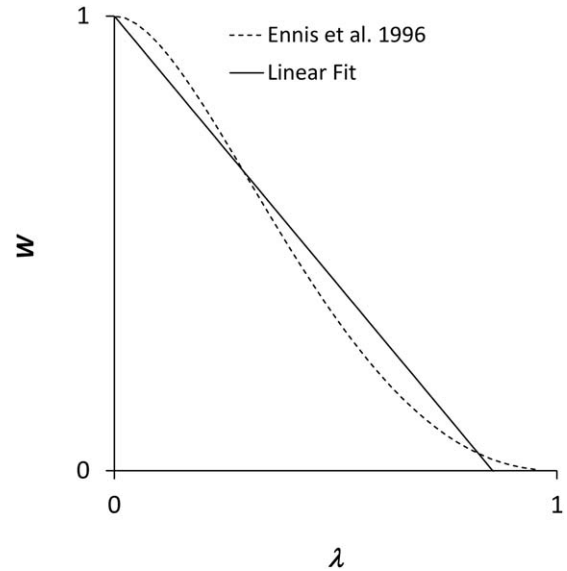


Figure 2. Comparison of the Ennis et al. equation for overall convective lag coefficient W to a linear approximation.

$$S = 1 - 1.17 \frac{r_s}{r_0} \frac{1}{\sqrt{1 - \frac{V}{V_{Max}}}} = 1 - \frac{1 - S_0}{\sqrt{1 - \frac{V}{V_{Max}}}} \quad (10)$$

Equation 10 is similar in form and function to the empirical model from Persson et al. (Eq. 5). It is notable that the flux can decline as the pore radius decreases and V approaches V_{Max} without loss of sieving in Eqs. 5 and 10 if the value of S_0 is near 1.

The predictions of the approximate equation for W (Eq. 10), the full model for W (Eq. 7), and the empirical model of Persson et al. (Eq. 5) are presented in Figure 3 assuming a typical S_0 value of 0.7 and calculating sieving from $0 \leq \lambda \leq 0.85$. The three models exhibit a concave shape, which is different than the convex shape exhibited by the numerical model of Filippov.⁴ The value of V/V_{Max} at the most fouled states were similar between the approximate model for W and the full model for W . At the point where λ reached 0.85, V/V_{Max} was 0.90 for the full model and 0.91 for the approximate model. If the two models were used to fit initial sieving, the approximate model would underpredict the final sieving values.

Equation 10 can be rearranged so the fitted parameter V_{Max} can be determined from a linear fit of volume and sieving data

$$V = V_{Max} \left[1 - \left(\frac{S_0 - 1}{S - 1} \right)^2 \right] \quad (11)$$

For constant flux experiments, pressure will rise as the membrane fouls, and Eq. 10 can be described as a function of time by substituting in $J_0 t$ for V . For constant pressure experiments, volume will increase with time according to the standard blocking law, which is applicable for the accumulation of foulant on the walls of straight cylindrical pores¹³

$$V = \left(\frac{1}{J_0 t} + \frac{1}{V_{Max}} \right)^{-1} \quad (12)$$

Substituting Eq. 12 into Eq. 10 yields an equation for sieving versus time at constant pressure

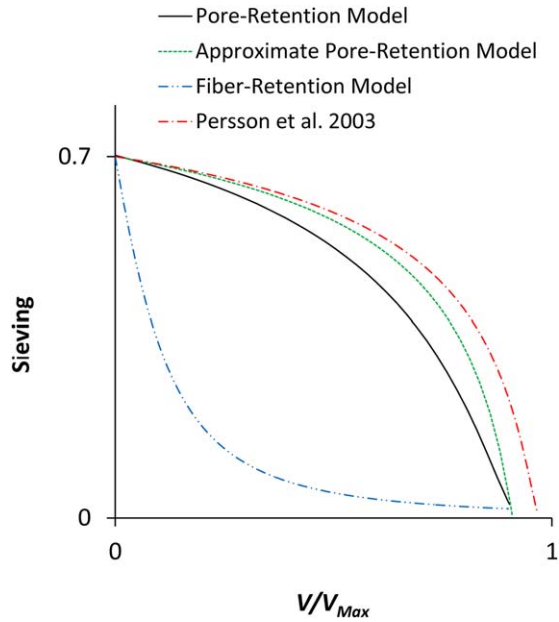


Figure 3. The sieving values predicted as a function of filtrate volume for the approximate equation for W (Eq. 10), the full equation for W (Eq. 7), the empirical model of Persson et al. (equation 2 in Persson et al., Eq. 5 here), and the fiber-retention model (Eq. 23) assuming a typical S_0 value of 0.7.

The pore-retention and Persson et al. models describe sieving decline due the adsorption of foulant on the walls of straight cylindrical membrane pores. The fiber-model describes sieving decline due to the adsorption of foulant on the walls of straight cylindrical fibers. The predictions of the model of Filippov et al. are similar to those of the fiber-model.

$$S = \frac{S_0 - 1}{\sqrt{1 + \frac{J_0 t}{V_{\text{Max}}}}} + 1 \quad (13)$$

Equation 13 can be rearranged so the fitted parameter V_{Max}/J_0 can be determined from a linear fit of time and sieving data

$$t = \frac{V_{\text{Max}}}{J_0} \left[\left(\frac{S-1}{S_0-1} \right)^2 - 1 \right] \quad (14)$$

Internal fouling of membranes composed of fibrous matrices

A model for sieving decline due to the adsorption of foulant on the walls of straight cylindrical fibers was generated. The model was developed by first relating the fiber radius to the volume of solids removed in a manner similar to that of the standard blocking model. A membrane with a fiber radius r_f , and total fiber length per area L_f will undergo an increase in fiber radius and solid fraction ϕ due to the filtration of a volume of a fouling liquid.¹⁷

$$L_f 2\pi r_f dr_f = \frac{c}{1-\epsilon} dV \quad (15)$$

Integrating Eq. 15 yields

$$\pi L_f (r_f^2 - r_{f0}^2) = \frac{c}{1-\epsilon} V \quad (16)$$

The filter solid fraction ϕ is related to the fiber radius, and filter thickness L_M

$$\phi = \frac{\pi r_f^2 L_f}{L_M} \quad (17)$$

$$\frac{\phi}{\phi_0} = \frac{r_f^2}{r_{f0}^2} \quad (18)$$

The solid fraction ϕ will be equal to 1 when the maximum volume V_{Max} has been filtered. Inserting Eq. 17 at $\phi=1$ into Eq. 16 yields

$$V_{\text{Max}} = L_M \frac{1-\epsilon}{c} (1-\phi_0) \quad (19)$$

Equation 17 can be arranged to solve for L_f in terms of the initial fiber radius and initial solid fraction. Inserting this into Eq. 16 yields the following, which can be rearranged by inserting Eq. 19 for V_{Max}

$$\frac{\phi}{\phi_0} = \frac{r_f^2}{r_{f0}^2} = 1 + \frac{c}{1-\epsilon} \frac{V}{\phi_0 L_M A} = 1 + \frac{1-\phi_0}{\phi_0} \frac{V}{V_{\text{Max}}} \quad (20)$$

A number of equations have been developed to describe the convective contribution to the transport of a spherical solute S through a fibrous matrix based on the ratio of the solute radius r_s to the fiber radius r_f and the volume fraction of fibers ϕ .

The sieving coefficient for the pure convective transport of globular proteins and Ficoll through agarose or acrylamide gels has been described

$$S = \left[1 + \beta \left[\frac{r_s}{r_f} \phi \right]^4 \right]^{-1} \quad (21)$$

where the value of β is either 2,400¹⁸ or 127.¹⁹ Substituting Eq. 18 for ϕ into Eq. 21 yields

$$S = \left[1 + \beta \left[\phi_0 \frac{r_s}{r_{f0}} \frac{r_f}{r_{f0}} \right]^4 \right]^{-1} \quad (22)$$

Substituting Eq. 20 for r_f/r_{f0} in terms of volume and initial porosity and assuming that when $V=0$ then $S=S_0$ yields the fiber-retention model, which describes sieving as a function of volume

$$S = \left[1 + \beta \left[\phi_0 \frac{r_s}{r_{f0}} \right]^4 \left[1 + \frac{1-\phi_0}{\phi_0} \frac{V}{V_{\text{Max}}} \right]^2 \right]^{-1} = \left[1 + \frac{1-S_0}{S_0} \left[1 + \frac{1-\phi_0}{\phi_0} \frac{V}{V_{\text{Max}}} \right]^2 \right]^{-1} \quad (23)$$

where the two fitted parameters are S_0 and $\phi_0 V_{\text{Max}}/1-\phi_0$.

The predictions of the fiber-retention model are presented in Figure 3 assuming a typical S_0 value of 0.7, ϕ_0 of 0.08, and calculating sieving from $0 \leq V/V_{\text{Max}} \leq 0.9$. The model exhibits a convex shape which is different than the predictions of the pore models. It is notable that the flux can decline as the fiber radius increases and V approaches V_{Max} without loss of sieving in Eq. 23 if the value of S_0 is near 1.

To obtain an equation for volume versus time at constant pressure, Equations 10 and 11 from Bolton et al.¹⁷ should be inserted into Eq. 23 for V

$$V = V_{\text{Max}} \left[1 + \frac{1}{\alpha} \left(1 - \left(1 + \frac{\alpha}{1-\phi_0} \right)^{\frac{1}{2}} \right) \right] \quad (24)$$

$$\alpha = \left(\frac{2J_0 t}{V_{\text{Max}}} + 1 \right) \frac{\phi_0}{1-\phi_0} - 1 \quad (25)$$

Equation 23 can be linearized in terms of sieving

$$V = \frac{\phi_0 V_{\text{Max}}}{1 - \phi_0} \left[\sqrt{\frac{1 - S}{1 - S_0}} - 1 \right] \quad (26)$$

Equation 23 was derived assuming that fibers never overlapped, and all solids deposited evenly along the fibers. It is possible, however, to have fibers that intersect or have a significant portion of their surface area in contact with one another. In this case the fiber radius will increase with permeate volume faster than that described by Eq. 16 because the solids will deposit on fiber surfaces only where there is no contact or overlap. This effect is more thoroughly evaluated elsewhere.¹⁷

Modeling of Sieving Decline Due to Fouling: External Membrane Fouling

Cake-growth: fouling due to surface cake layer with increasing thickness

Cake fouling refers to an increase in flow resistance and/or a decline in sieving due to a layer of foulant that builds up uniformly on top of a membrane. The sieving coefficient S for a membrane covered in a cake will be equal to the multiple of the membrane, S_M , and cake, S_C , sieving coefficients: $S = S_M S_C$. This relationship is applicable assuming the retentate is dilute.²⁰

The cake layer was assumed to be composed of an incompressible secondary membrane of increasing thickness. The cake sieving coefficient S_C depends on the overall hindrance factors for convection W and diffusion H , the protein free solution diffusivity, D , the cake height h , and the solvent velocity J according to Eq. 6

$$S_C = \frac{W}{1 - (1 - W) \exp\left(-\frac{WhJ}{HD}\right)} \quad (27)$$

As the membrane fouls, the cake will increase in thickness and sieving will decline. At $V = \infty$ the membrane thickness will prevent any diffusive contributions to sieving, the exponential term in Eq. 27 will be equal to zero, and S_C will be equal to the $S_{C\infty}$. Inserting $S_{C\infty}$ into Eq. 27 and rearranging yields

$$S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp\left(-\frac{S_{C\infty} h J}{HD}\right)} \quad (28)$$

In practice, if the membrane is not retentive to the protein but the cake layer is, then $S_{C\infty}$ will be equal to the measured final sieving value. Alternatively, $S_{C\infty}$ can be fit to the data. The height h of an incompressible external cake of constant morphology will increase linearly with volume filtered V .²¹ Defining h in terms of volume and a new cake-growth parameter K_{CG} yields

$$h = K_{CG} V \frac{HD}{S_{C\infty} J_0} \quad (29)$$

Inserting for h into Eq. 28 yields

$$S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp\left(-K_{CG} V \frac{J}{J_0}\right)} \quad (30)$$

It is notable that the sieving in Eq. 30 depends not just on the volume filtered, as in other models, but on the solvent velocity J . This is because the diffusive contribution to sieving will be higher at lower solvent velocities.

If the flow rate J is kept constant at the initial flow rate J_0 , Eq. 30 simplifies to

$$S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp(-K_{CG} V)} \quad (31)$$

At constant pressure, the volume and solvent velocity can be calculated as a function of time using the cake-fouling model and the fitted parameter K_C ¹³

$$V = \frac{\sqrt{1 + 2K_C J_0^2 t} - 1}{K_C J_0} \quad (32)$$

$$\frac{J}{J_0} = \frac{1}{1 + K_C J_0 V} = \frac{1}{\sqrt{1 + 2K_C J_0^2 t}} \quad (33)$$

Inserting Eqs. 32 and 33 into Eq. 30 provides an equation for sieving as a function of time

$$S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp\left(-\frac{K_{CG}}{K_C J_0} \left[1 - \frac{1}{\sqrt{1 + 2K_C J_0^2 t}}\right]\right)} \quad (34)$$

It is notable that the cake-growth model predicts that the sieving would decline initially as the cake layer thickness increases and the diffusive contribution to sieving declines to zero. Subsequently, however, the sieving would not decline despite the growing cake layer. The resistance to solvent flow would increase with increasing cake thickness and be observed as a decline in flux when operating at constant pressure, or an increase in pressure when operating at constant flow. The predicted sieving however would remain constant.

This is illustrated in Figure 4, where the sieving can either decline to near zero or decline to some intermediate value and then remain constant depending on the values of K_{CG} and $S_{C\infty}$.

Cake-series: fouling due an increasing number of cake layers in series

The sieving coefficient S for a membrane covered in a cake will be equal to the multiple of the membrane, S_M , and cake, S_C , sieving coefficients: $S = S_M S_C$. If a number n of discrete cake layers deposit on top of one another on the membrane, then the cake sieving coefficient will be equal to the multiple of the sieving coefficients of the individual layers. Assuming that the sieving coefficient of each cake layer, S_{CL} , is equal yields

$$S_C = S_{CL1} S_{CL2} \dots S_{CLn} = S_{CL}^n \quad (35)$$

Equation 35 is valid assuming a dilute retentate.²⁰ If it is assumed that the number of discrete cake layers n increases in a smooth linear fashion with the volume filtered V then Eq. 35 can be written in terms of a constant F , assuming $n = FV$, and then rearranged and put in terms of the cake-series constant $K_{CS} = -F \ln S_{CL}$ to derive the cake-series sieving model

$$S_C = S_{CL}^{FV} = S_{CL}^{\text{FV}} = \exp(FV \ln S_{CL}) = \exp(-K_{CS} V) \quad (36)$$

At constant pressure, Eq. 32 can be inserted for volume into Eq. 36 to calculate sieving as a function of time

$$S_C = \exp\left(\frac{-K_{CS}}{K_C J_0} \left[\sqrt{1 + 2K_C J_0^2 t} - 1\right]\right) \quad (37)$$

The sieving versus volume predicted by the cake-series model is compared to the predictions of the cake-growth

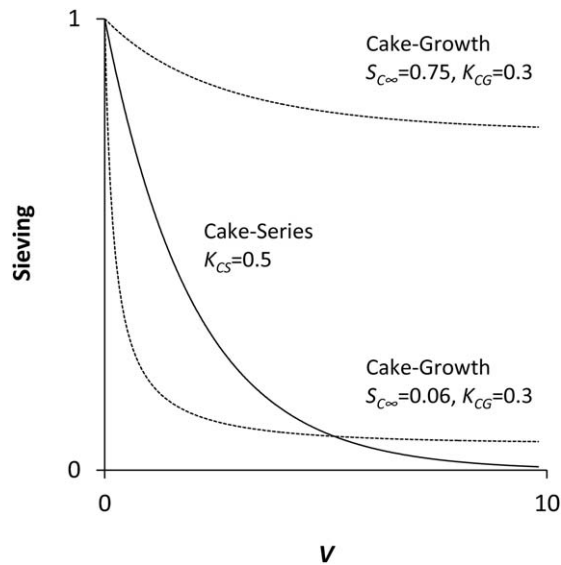


Figure 4. Predictions of the cake-growth and cake-series models as a function of volume filtered and fitted parameter values.

The cake-growth model describes sieving decline due to the increasing thickness of a surface cake layer. The cake-series model describes sieving decline due to an increasing number of surface cake layers in series.

model in Figure 4. Both models have similar convex behavior that resembles that of the fiber-coating model in Figure 3. The cake-series model predicts sieving will steadily decline and approach zero, while the cake-growth model has an additional fitted parameter and predicts sieving will decline and then remain constant at $S_{C\infty}$, where $S_{C\infty}$ can vary between near 0 and 1. This behavior has been observed in a number of studies in the literature including in figure 3 of Blanpain-Avet et al.,² figure 13 of Guell and Davis,⁸ figure 4 of Kuberkar and Davis,⁹ and figure 3 of Persson et al.¹² The cake-growth model predictions depend on the volume filtered V and the solvent flux J , while the cake-series model predictions only depend on the volume filtered.

Modeling of Sieving Decline Due to Fouling: Combined Internal and External Membrane Fouling

It is possible to combine the models for internal and external fouling to obtain models that describe both mechanisms occurring simultaneously or sequentially. The sieving coefficient S for a membrane covered in a cake will be equal to the multiple of the membrane, S_M , and cake, S_C , sieving coefficients: $S = S_M S_C$. This relationship is applicable assuming the retentate is dilute.²⁰

The pore-retention or fiber-retention models can be used for calculation of S_M . The cake-series or cake-growth models can be used for calculation of S_C .

The pore-retention model can be multiplied by the cake-growth model to obtain equations for sieving as a function of volume or time assuming the two fouling mechanisms are occurring simultaneously

$$S = \left[1 - \frac{1-S_0}{\sqrt{1-\frac{V}{V_{\text{Max}}}}} \right] \frac{S_{C\infty}}{1 - (1-S_{C\infty})\exp(-K_{CG}V)} \quad (38)$$

$$S = \left[1 - \frac{1-S_0}{\sqrt{1-\frac{J_0 t}{1+\frac{J_0 t}{V_{\text{Max}}}}}} \right] \frac{S_{C\infty}}{1 - (1-S_{C\infty})\exp\left(-\frac{K_{CG}}{K_C J_0} \left[1 - \frac{1}{\sqrt{1+2K_C J_0^2 t}} \right] \right)} \quad (39)$$

The pore-retention model can be combined with the cake-series model to obtain equations for sieving as a function of volume or time

$$S = \left[1 - \frac{1-S_0}{\sqrt{1-\frac{V}{V_{\text{Max}}}}} \right] \exp(-K_{CS}V) \quad (40)$$

$$S = \left[1 - \frac{1-S_0}{\sqrt{1-\frac{J_0 t}{1+\frac{J_0 t}{V_{\text{Max}}}}}} \right] \exp\left(\frac{-K_{CS}}{K_C J_0} \left[\sqrt{1+2K_C J_0^2 t} - 1 \right] \right) \quad (41)$$

It is mathematically simple to combine internal fiber fouling with caking. However, the fiber-coating model and the two caking models all have convex shapes and combining them does not provide advantages. It is likely a good fit of a dataset could be obtained assuming either internal fiber fouling or external caking.

The combined caking and internal pore plugging models are very powerful in that they have the ability to fit sieving data that declines in a convex manner, a concave manner, or any combination of the two. For example, these models have the capability of fitting a linear decline of sieving with volume filtered.

It is possible to combine the models for internal and external fouling to obtain models that describe the two mechanisms occurring sequentially. As is described below, internal blocking commonly occurs initially as foulants enter the pores and then caking occurs later when foulants are excluded from the constricted pores. To generate a single equation for the two mechanisms occurring sequentially, a mathematical method was sought to effectively turn off one mechanism and turn on another mechanism at some intermediate transition time, t_T , during the run. The logistic function, $1/(1+\exp[z(t-t_T)])$ was used to smoothly simulate a step function that increases rapidly from zero before t_T to 1 after t_T .

The time t in Eqs. 39 and 41 was broken up into a pore-retention time, t_P , which occurs before fouling transitions from pore blocking to caking at $t = t_T$, and a caking time, t_C , which occurs after caking begins. The caking time t_C and pore-retention time t_P were calculated using the logistic function, where the constant z was arbitrarily set to a high value of $200/t_T$ to ensure a rapid transition regardless of the timescale used

$$t_C = \frac{t - t_T}{1 + \exp\left[200\left(\frac{t - t_T}{t_T}\right)\right]} \quad (42)$$

$$t_P = t - t_C \quad (43)$$

The high value of z ensures it does not impact the model fit. The pore-retention time t_P will be equal to t until the fouling transitions to caking at $t = t_T$. Thereafter t_P will remain constant at $t_P = t_T$ and no further sieving decline due to pore-retention will occur. The caking time t_C will be equal to 0 and no sieving decline due to cake-growth will occur until the fouling transitions to caking at $t = t_T$. Thereafter t_C will increase and be equal to $t - t_T$.

The pore-retention time t_p is substituted for t in the left sides of Eqs. 39 and 41 and the caking time t_c is substituted for t in the right sides of Eqs. 39 and 41 to generate the new sequential models.

$$S = \left[1 - \frac{1-S_0}{\sqrt{1 + \frac{J_0^2}{V_{\text{Max}}^2}}} \right] \frac{S_{C\infty}}{1 - (1-S_{C\infty}) \exp\left(-\frac{K_{CG}}{K_C J_0} \left[1 - \frac{1}{\sqrt{1 + 2K_C J_0^2 t_c}} \right] \right)} \quad (44)$$

$$S = \left[1 - \frac{1-S_0}{\sqrt{1 + \frac{J_0^2}{V_{\text{Max}}^2}}} \right] \exp\left(\frac{-K_{CS}}{K_C J_0} \left[\sqrt{1 + 2K_C J_0^2 t_c} - 1 \right] \right) \quad (45)$$

To derive equations for sequential fouling as a function of volume, a transition volume V_T is used

$$V_C = \frac{V - V_T}{1 + \exp\left[200\left(\frac{V - V_T}{V_T}\right)\right]} \quad (46)$$

$$V_P = V - V_C \quad (47)$$

The pore-retention volume V_P will be equal to V until the fouling transitions to caking at $V = V_T$. Thereafter, V_P will remain constant at $V_P = V_T$. The caking time V_C will be equal to 0 until the fouling transitions to caking at $V = V_T$. Thereafter, V_C will increase and be equal to $V - V_T$.

The pore-retention volume V_P will be substituted for V in the left sides of Eqs. 38 and 40 and the caking volume V_C is substituted for V in the right sides of Eqs. 38 and 40 to generate the new sequential models.

$$S = \left[1 - \frac{1-S_0}{\sqrt{1 - \frac{V_P}{V_{\text{Max}}}}} \right] \frac{S_{C\infty}}{1 - (1-S_{C\infty}) \exp(-K_{CG} V_C)} \quad (48)$$

$$S = \left[1 - \frac{1-S_0}{\sqrt{1 - \frac{V_P}{V_{\text{Max}}}}} \right] \exp(-K_{CS} V_C) \quad (49)$$

A summary of all models derived in this study is provided in Table 1.

Application of Models to Experimental Sieving Data

Filtration of beer through track-etched MF

A well-documented dataset that provided both sieving and membrane resistance as a function of time or volume was sought to test the models. Data for the microfiltration of clarified beer using a membrane with straight cylindrical pores were used to test the models. Although this system is different than the microfiltration of a bioreactor, the complexity of the feed stream, the ideal nature of the membrane pores, and the extensive documentation make it valuable for use in this study. The data in figure 8 of Blanpain-Avet et al.² shows standard blocking occurring initially, based on the steepness of the flow resistance profile, and cake fouling occurring subsequently. The pore-retention and cake-growth models developed in this study were used sequentially and found to provide good fits of these data.

The Blanpain-Avet et al. data were obtained using a lager beer that had been clarified using a diatomaceous earth sheet and had less than 5 yeast cells per 100 ml. Sieving and

membrane resistance were measured during microfiltration of the clarified beer through 12.5 cm² membranes containing track-etched straight cylindrical 0.2 micron pores. Experiments were done in a 200 mL stirred cell using 850 RPM and 100 kPa.

The membrane flow resistance initially increased rapidly due to standard pore blocking as foulants were able to enter and foul the pores. These initial data were well described by the standard blocking model, $R/R_0 = (1 + J_0 t / V_{\text{Max}})^2$, using a J_0/V_{Max} value of 1.30 s⁻¹, as shown in Figure 5. After about 28 min, the membrane resistance increased more slowly due to caking as foulants built up on the membrane surface. The data after 28 min was well described by the cake blocking model, $R/R_0 = (1 + 2K_C J_0^2 t)^{1/2}$, using a $K_C J_0^2$ value of 3.03×10^{-4} s⁻¹, as shown in Figure 5.

Sieving data measured using both the Bradford and Lowry protein assays were provided in Blanpain-Avet et al. figure 8b. The Bradford assay data, which showed higher sieving loss, were used to fit the sequential pore-retention and cake-growth model in this study. Similar fits were obtained with the Lowry data (not shown). In addition, similar fits were obtained using the data obtained at lower pressure in figure 8a (not shown).

Data are provided for the experimentally observed sieving coefficient (S_{Obs}), which is equal to the ratio of the bulk retentate concentration to the filtrate concentration. However, due to concentration polarization, the concentration at the membrane surface is higher than the bulk retentate concentration. When modeling membrane sieving, typically the intrinsic sieving coefficient (S_{Int}), which is equal to the ratio of the concentration at the membrane surface to the filtrate concentration, is calculated from the observed sieving coefficient using the system dimensions, protein diffusivity, and mass transfer coefficient. In this case, the sieving data were not corrected for concentration polarization because the system dimensions were not defined adequately for calculation of a mass transfer coefficient.

The sieving data during the first 28 min of filtration, where pore plugging was occurring, declined in a concave fashion, similar to the predictions of the pore-retention model. The sieving data after the first 28 min of filtration, where cake fouling was occurring, declined in a convex fashion and then remained constant, similar to the predictions of the cake-growth model. The sequential pore-retention and cake-growth model (Eq. 44) was fit to the data using two fitted parameters and three independent parameters. The model used the J_0/V_{Max} (1.30 s⁻¹), the $K_C J_0^2$ (3.03×10^{-4} s⁻¹), and the t_T (28 min) determined independently above from the resistance data. The measured final sieving coefficient value, $S = 0.52$, which is equal to $S_M S_{C\infty}$, was used with the measured sieving value at the end of pore plugging, $S_M = 0.85$, to determine $S_{C\infty} = 0.615$.

The model was fit by varying the value of S_0 and $K_{CG}/K_C J_0$ to minimize the sum of the squares of the error between the model predictions and the sieving data. The S_0 value that provided the best fit, 0.977, was similar to the measured initial sieving coefficient: 0.95. The parameter values and error of the fits of the membrane resistance and sieving models to the data are shown in Table 2.

The models fit the data well, as shown in Figure 5, considering only two fitted parameters and four independently determined parameters were used.

Table 1. Summary of the Models of Sieving Loss as a Function of Membrane Fouling

Model	Model of sieving versus volume	Model of sieving versus time at constant pressure	Model parameters
Pore-retention	$S = 1 - \frac{1 - \delta_0}{\sqrt{1 - \frac{V}{V_{\text{Max}}}}}$	$S = 1 - \frac{1 - \delta_0}{\sqrt{1 + \frac{t}{t_{\text{Max}}}}}$	V_{Max} (m), δ_0
Fiber-retention	$S = \left[1 + \frac{1 - \delta_0}{\delta_0} \left[1 + \frac{1 - \phi_0}{\phi_0} \frac{V}{V_{\text{Max}}} \right]^2 \right]^{-1}$	$S = \left[1 + \frac{1 - \delta_0}{\delta_0} \left[1 + \frac{1 - \phi_0}{\phi_0} \left[1 + \frac{1}{\alpha} \left(1 - \left(1 + \frac{\alpha}{1 - \phi_0} \right)^{\frac{1}{\alpha}} \right) \right]^2 \right] \right]^{-1}$	V_{Max} (m ⁻¹), K_C (s m ⁻²), S_0 , ϕ_0
Cake-growth	$S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp(-K_{CG} V_0^{\frac{1}{2}})}$	$\alpha = \left(\frac{2\phi_0}{V_{\text{Max}}} + 1 \right) \frac{\phi_0}{1 - \phi_0} - 1$ $S_C = \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp\left(-\frac{K_{CG}}{K_{CG}'} \left[1 - \frac{1}{\sqrt{1 + 2K_C \phi_0^2}} \right]\right)}$	K_{CG} (m ⁻¹), K_C (s m ⁻²), $S_{C\infty}$
Cake-series	$S_C = \exp(-K_{CS} V)$	$S_C = \exp\left(-\frac{K_{CS}}{K_{CG}'} \left[\sqrt{1 + 2K_C \phi_0^2} - 1 \right]\right)$	K_{CS} (m ⁻¹), K_C (s m ⁻²)
Simultaneous pore-retention and cake-growth	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 - \frac{V}{V_{\text{Max}}}}} \right] \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp(-K_{CG} V)}$	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 + \frac{t}{t_{\text{Max}}}}} \right] \frac{S_{C\infty}}{\sqrt{1 + \frac{t}{t_{\text{Max}}}}} *$	V_{Max} (m), K_{CG} (m ⁻¹), K_C (s m ⁻²), $S_{C\infty}$, S_0
Sequential pore-retention followed by cake-growth	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 - \frac{V_P}{V_{\text{Max}}}}} \right] \frac{S_{C\infty}}{1 - (1 - S_{C\infty}) \exp(-K_{CG} V_C)}$ $V_C = \frac{V - V_T}{1 + \exp\left[200 \left(\frac{V - V_T}{V_T}\right)\right]}$ $V_P = V - V_C$	$1 - (1 - S_{C\infty}) \exp\left(-\frac{K_{CG}}{K_{CG}'} \left[1 - \frac{1}{\sqrt{1 + 2K_C \phi_0^2}} \right]\right)$ $S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 + \frac{t_P}{t_{\text{Max}}}}} \right] \frac{S_{C\infty}}{\sqrt{1 + \frac{t_P}{t_{\text{Max}}}}} *$ $1 - (1 - S_{C\infty}) \exp\left(-\frac{K_{CG}}{K_{CG}'} \left[1 - \frac{1}{\sqrt{1 + 2K_C \phi_0^2}} \right]\right)$ $t_C = \frac{t - t_P}{1 + \exp\left[200 \left(\frac{t - t_P}{t_P}\right)\right]}$ $t_P = t - t_C$	V_{Max} (m), K_{CG} (m ⁻¹), K_C (s m ⁻²), $S_{C\infty}$, S_0 , V_T (m), t_T (s)
Simultaneous pore-retention and cake-series	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 - \frac{V}{V_{\text{Max}}}}} \right] \exp(-K_{CS} V)$	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 + \frac{t}{t_{\text{Max}}}}} \right] \exp\left(-\frac{K_{CS}}{K_{CG}'} \left[\sqrt{1 + 2K_C \phi_0^2} - 1 \right]\right)$	V_{Max} (m), K_{CS} (m ⁻¹), K_C (s m ⁻²), S_0
Sequential pore-retention followed by cake-series	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 - \frac{V_P}{V_{\text{Max}}}}} \right] \exp(-K_{CS} V_C)$ $V_C = \frac{V - V_T}{1 + \exp\left[200 \left(\frac{V - V_T}{V_T}\right)\right]}$ $V_P = V - V_C$	$S = \left[1 - \frac{1 - \delta_0}{\sqrt{1 + \frac{t_P}{t_{\text{Max}}}}} \right] \exp\left(-\frac{K_{CS}}{K_{CG}'} \left[\sqrt{1 + 2K_C \phi_0^2} - 1 \right]\right)$ $t_C = \frac{t - t_P}{1 + \exp\left[200 \left(\frac{t - t_P}{t_P}\right)\right]}$ $t_P = t - t_C$	V_{Max} (m), K_{CS} (m ⁻¹), K_C (s m ⁻²), S_0 , V_T (m), t_T (s)

mAb harvest using TFF MF

Data from a well-characterized literature study describing the loss of antibody sieving over the course of TFF microfiltration of a perfusion cell culture were used to test the models developed in this study. The cake-series and cake-growth models were found to provide good fits of the datasets “TFF10” and “TFF6” in figure 7a of Clincke et al.¹⁰ The sieving data in these datasets declined with filtration time in a convex fashion, similar to the model predictions.

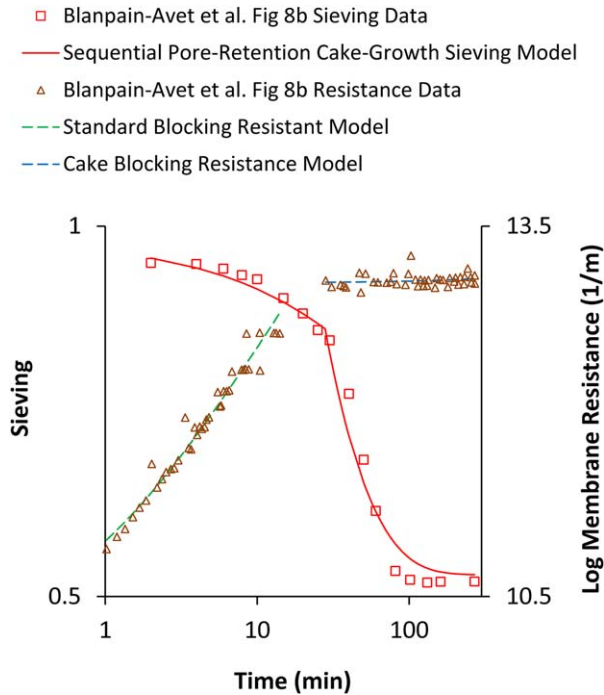


Figure 5. Fit of the membrane resistance and sieving data presented in figure 8b of Blanpain-Avet et al. to flow models and the new sequential pore-retention followed by cake-growth model developed in this study.

The standard blocking model ($t < 28$ min) and the cake fouling model ($t > 28$ min) provided good fits of the resistance data. The sequential pore-retention followed by cake-growth model provided good fits of the sieving data using J_0/V_{Max} , t_T , and $K_C J_0^2$ from the resistance models, $S_{C\infty}$ from the sieving data, and fitting $K_{CG}/K_C J_0$ and S_0 to the model.

In the Clincke et al. study, a CHO cell bioreactor containing 0–500 mg L⁻¹ IgG was filtered through a 850 cm² hollow fiber device containing 50 fibers at constant flow for about 20 days using a microfiltration system. The fibers had 0.2 μm pores and were 60 cm long with 1 mm lumen. The permeate flux v was 8.17×10^{-7} m s⁻¹ for TFF10 and 6.54×10^{-7} for TFF6. The feed velocity v_F was 0.127 m s⁻¹ for TFF6. The feed velocity v_F was 0.297 m s⁻¹ for days 1 through 7 and 0.424 m s⁻¹ after day 7 for the TFF10 experiment.

The sieving data presented in Clincke et al. was corrected for concentration polarization at the membrane surface before fitting in this study. The intrinsic sieving coefficient (S_{Int}) was calculated from the observed sieving coefficient (S_{Obs}) data using the following equation,²² where v is the permeate flux and k is the mass transfer coefficient:

$$S_{\text{Int}} = \frac{S_{\text{Obs}}}{(1 - S_{\text{Obs}}) \exp\left(\frac{v}{k}\right) + S_{\text{Obs}}} \quad (50)$$

The mass transfer coefficient k was calculated as a function of feed velocity v_F , fiber length L_{HF} , fiber diameter d_{HF} , and IgG free solution diffusivity D (4×10^{-11} m² s⁻¹)²³ using the Leveque solution for the Sherwood number Sh for flow in hollow fibers²⁴:

$$Sh = \frac{k d_{\text{HF}}}{D} = 1.62 \left(\text{Re} Sc \frac{d_{\text{HF}}}{L_{\text{HF}}} \right)^{1/3} = 1.62 \left(\frac{v_F d_{\text{HF}}^2}{D L_{\text{HF}}} \right)^{1/3} \quad (51)$$

The intrinsic sieving coefficient was 0–38% lower than the observed sieving coefficient for the two data sets. The difference was largest for the late data points. Based on their figure 3h¹¹ the perfusion pump began pumping on day 3.68 or 1.56 for runs TFF6 and TFF10, respectively. Therefore, these time values were subtracted from the time data from Clincke et al figure 7a before fitting the cake-growth or cake-series models.

The sieving data were fit using the cake-series and cake-growth models by varying the fitted parameters to minimize the sum of the squares of the error between the model predictions and the sieving data. The fitted parameters and model fit errors are shown in Table 3. The sieving data and model predictions are compared in Figures 6 and 7.

The fit of the cake-growth and cake-series models provide physical insights into the fouling mechanisms. Both models accurately fit the convex shape of the sieving decline data

Table 2. Summary of the Resistance and Sieving Model Fitting of the Data from Figure 8b of Blanpain-Avet et al.

Model	Interval	Model fit error, SSR	Fit parameters	Input parameters
Standard blocking	$t < 28$ min	2.37×10^{-3}	$J_0/V_{\text{Max}} = 1.30 \text{ s}^{-1}$	$J_0/V_{\text{Max}} = 1.30 \text{ s}^{-1}$, $K_C J_0^2 = 3.03 \times 10^{-4} \text{ s}^{-1}$, $S_{C\infty} = 0.615$, $t_T = 28$ min
Cake filtration	$t > 28$ min	5.90×10^{-4}	$K_C J_0^2 = 3.03 \times 10^{-4} \text{ s}^{-1}$	
Sequential pore-retention cake-growth	0–271 min	1.19×10^{-2}	$S_0 = 0.977$, $K_{CG}/K_C J_0 = 106$	

The standard blocking and cake filtration models were fit to the membrane flow resistance data. The sequential pore-retention followed by cake-growth model (Eq. 44) was fit to the sieving data

Table 3. Summary of the Cake-Series (Eq. 36) and Cake-Growth (Eq. 31) Model Fitting of the TFF6 and TFF10 Data from Figure 7a of Clincke et al.

Model	Data Set	Model fit error, SSR	Fit parameter values
Cake-growth	TFF6	0.33	$K_{CG} = 1.05 \times 10^{-3} \text{ m}^{-1}$, $S_{\infty} = 1.72 \times 10^{-4}$
Cake-series	TFF6	1.00	$K_{CS} = 2.49 \text{ m}^{-1}$
Cake-growth	TFF10	0.69	$K_{CG} = 2.34 \times 10^{-4} \text{ m}^{-1}$, $S_{\infty} = 2.91 \times 10^{-5}$
Cake-series	TFF10	1.85	$K_{CS} = 2.43 \text{ m}^{-1}$

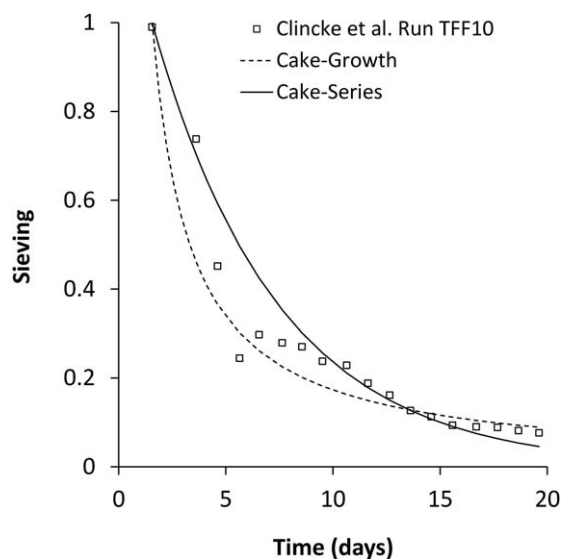


Figure 6. Fit of the cake-series and cake-growth models to the TFF10 sieving data from Clincke et al.

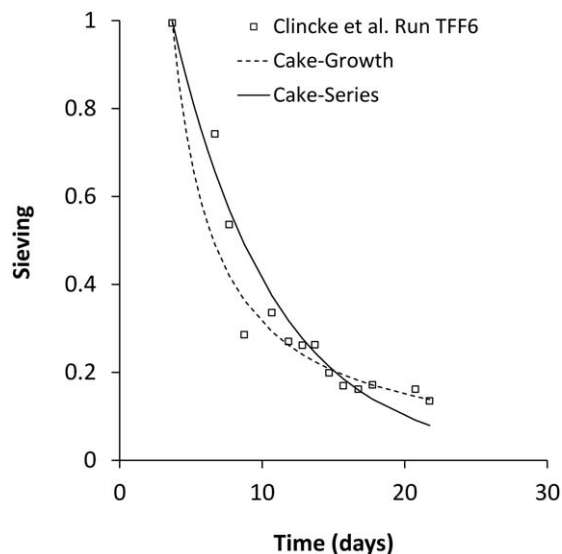


Figure 7. Fit of the cake-series and cake-growth models to the TFF6 sieving data from Clincke et al.

(Figures 6 and 7). The cake-growth model provided lower model fit error, although it benefits from the use of two fitted parameters, compared to a single fitted parameter for the cake-series model. The fit of the cake-growth and cake-series models suggest that fouling was due to external caking and not due to internal pore fouling (pore-retention). The pore-retention model predicts a concave-shaped sieving decline.

Conclusions

Variable or declining product sieving can impair process robustness and economics. In this study, new explicit mathematical models were generated to describe the loss of protein sieving as a function of membrane fouling.

Two models of sieving loss due to internal membrane fouling were generated that explicitly treated membrane structure, solute partitioning, and hindered diffusion and convection of solutes. The two-parameter pore-retention model of sieving decline in membranes with straight cylindrical

pores due to the accumulation of foulant on pore walls was generated. A simplified approximate model that can be linearized was generated to simplify calculation. The two-parameter fiber-retention model of sieving decline in fibrous membranes due to the accumulation of foulant on fiber surfaces was generated. Both sieving models incorporated the same parameters as those used in the corresponding flux decay models. This allows parameters to be determined either by flux or sieving decay data. It is notable that for both models it is predicted that flux can decline without loss of sieving if the initial sieving value is near 1.

Two models of sieving loss due to external fouling were generated. The one-parameter cake-series model of sieving decline as a function of volume filtered due to the accumulation of a series of incompressible cake layers was generated. The sieving is predicted to decline steadily to zero. The two-parameter cake-growth model of sieving decline as a function of volume filtered due to the accumulation of an incompressible cake layer of increasing thickness was generated. Sieving predicted by the model depends not just on the volume filtered but on the solvent velocity J due to diffusive contributions to sieving. The cake-growth model is well suited for data sets where sieving declines initially and then stabilizes at some intermediate value, despite a continued flow resistance increase, as has been observed in literature.

The fiber-retention, cake-series, and cake-growth models predict that sieving will decline with volume in a convex fashion, while the pore-retention model predicts a concave behavior.

The pore-retention model was combined with either the cake-series or cake-growth models to obtain models that describe internal and external fouling occurring either simultaneously or sequentially. The combined models can fit convex, concave and linear sieving decline curves.

The sequential pore-retention followed by cake-growth model developed in this study was applied to data for the microfiltration of clarified beer using straight cylindrical pores membranes. Based on the flow resistance profile, standard blocking occurred initially and caking occurred subsequently. The model was found to provide good fits of both the early pore-retention sieving decline and also the late cake-growth sieving decline.

The cake-series and cake-growth models were found to provide similar good fits of literature data describing the loss of antibody sieving over the course of microfiltration of a perfusion cell culture.

The new models provide insights into the internal and external mechanisms of fouling that result in the loss of product sieving. The models can be used to fit sieving data to determine the mechanisms underlying sieving loss, to extrapolate sieving data beyond experimental filtration volumes, and to predict the effects of changing operational parameters.

Notation

- A = available membrane frontal area, m^2
- c = volume concentration of foulant
- D = free solution diffusivity, $\text{m}^2 \text{s}^{-1}$
- d_{HF} = hollow fiber diameter, m
- F = cake series model placeholder, $\text{m}^2 \text{m}^{-3}$
- h = height of an incompressible cake, m
- H = overall hindrance factors for diffusion
- J = flux, m s^{-1}
- K_C = cake fouling model constant, s m^{-2}

K_{CS} = cake-series constant, m^{-1}
 K_{CG} = cake-growth constant, m^{-1}
 k = mass transfer coefficient, $m\ s^{-1}$
 L_M = filter thickness, m
 L_{HF} = hollow fiber length, m
 n = number of discrete cake layers
 N = pore density, m^{-2}
 L_f = fiber length per area, $m\ m^{-2}$
 r = pore radius, m
 r_f = fiber radius, m
 r_s = solute radius, m
 S_M = membrane sieving coefficient
 S_C = cake sieving coefficient
 S_{Int} = intrinsic sieving coefficient
 S_{Obs} = observed sieving coefficient
 $S_{C\infty}$ = sieving coefficient of cake at infinite time
 S_{CL} = sieving coefficient of each cake layer
 t = time, s
 t_C = caking time, s
 t_P = pore-retention time, s
 t_T = time at transition from pore-retention to caking, s
 V = normalized volume filtered, $m^3\ m^{-2}$
 V_C = caking volume, $m^3\ m^{-2}$
 v_F = feed velocity, $m\ s^{-1}$
 V_{Max} = volume filtered when the porosity declines to zero, $m^3\ m^{-2}$
 V_P = pore-retention volume, $m^3\ m^{-2}$
 V_T = volume at transition from pore-retention to caking, $m^3\ m^{-2}$
 W = overall hindrance factor for convection
 z = exponential factor in logistic equation

Greek letters

α = placeholder in fiber coating model
 β = placeholder in fiber coating model
 ϵ = porosity of foulant
 λ = ratio of the solute radius r_s to the pore radius r
 ϕ = volume fraction of fibers

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